

Before starting, make sure you **read all the questions** and understand what is asked of you. Write your answers to the questions in a separate sheet of paper and make sure you **justify all of your answers**. Multiple-choice questions have a **unique correct answer**.

1. (5 points) Which one of the following is an advantage of automata-based scanners over hand-built scanners?
 - A. They can always be automatically constructed.
 - B. They can go beyond regular languages.
 - C. They implicitly factor out common prefixes.
 - D. They require less coding effort.
2. (5 points) If you are implementing an LL(k) parser based on a given grammar, which version of recursion should you favor when manipulating/rewriting the grammar?
 - A. Left recursion
 - B. Right recursion
3. (5 points) Which one of the following languages is not a regular language?
 - A. The set of all words on $\Sigma = \{[,], (,)\}$ such that every $)$ matches a previous $[$ and every $($ is eventually matched to a $]$.
 - B. The set of all prefixes of the string “Compilers is the best course in our program”.
 - C. The set of all first names of people born in 2000.
 - D. The set of all French words.
4. (15 points) Consider the language of (integer) arithmetic expressions defined by the following grammar.

(1)	S	$\rightarrow$	Exp
(2)	Exp	$\rightarrow$	$\text{Exp} * \text{Exp}$
(3)	Exp	$\rightarrow$	$\text{Exp} + \text{Exp}$
(4)	Exp	$\rightarrow$	cst

- (a) (5 points) Rewrite the grammar so that all parse trees respect the order of operations (i.e. multiplication before addition).
 - (b) (10 points) Is the language LL(1)?
5. (15 points) Consider once more the language of arithmetic expressions and add to it parentheses (with the usual interpretation that they have higher precedence than all other operations).
 - (a) (5 points) Give the grammar for the new language, again so that all parse trees respect the order of operations.
 - (b) (10 points) Is the language LL(1)?

Python dictionaries

For the questions below, we will focus on the language of (simplified) Python dictionaries. A Python dictionary is a **value**. In turn, a value can be one of the following:

- a **dictionary**,
- a (possibly negative) integer number,
- a constant from true, false, or null.

Then, a dictionary is a curly-brace surrounded, comma-separated sequence of pairs $k : v$ where k is a **key** (a C-like identifier) and v is a value.

6. (5 points) Is the language of Python dictionaries a regular language? Why or why not?
7. (10 points) (a) (5 points) Write regular expressions for lexical units for identifiers, for numbers, and for all three constants.
(b) (5 points) Give minimal deterministic finite state automata for all 5 languages too.
8. (10 points) Suppose we are given a **deterministic finite automaton** for a language $\mathcal{L}$ on Σ . For all words $w \in \Sigma^*$, we can use the automaton to determine whether $w \in \mathcal{L}$ in time:
 - A. constant in $|w|$
 - B. nope, that is an undecidable problem
 - C. linear in $|w|$
9. (5 points) Argue that the following language over the alphabet $\Sigma = \{0, 1\}$ is context free:

$$L_{\text{pal}} = \{ww^R \mid w \in \Sigma^*\}$$

where w^R denotes the word spelled in reverse order.

10. (10 points) Argue that L_{pal} , from the previous question, cannot be recognized by a deterministic pushdown automaton.
11. (10 points) Give an example of an LR(1) grammar that is not LALR(1).
12. (5 points) Give an example of a context-free language that is not LR(1).